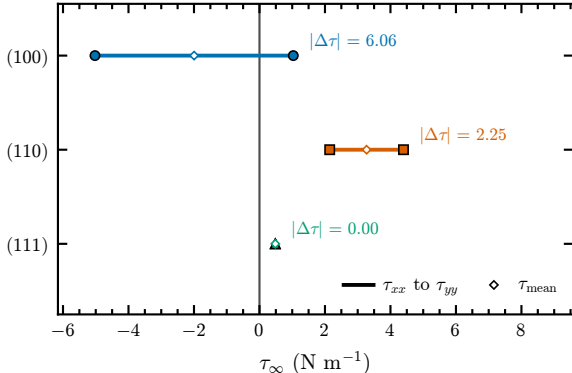




Extrapolated surface stress of each H-terminated facet. Each bar runs from τ_{xx} to τ_{yy} for one surface, with τ_{mean} as the open diamond between them and error bars giving the rms of the thickness extrapolation (at most 0.029 N/m, smaller than the markers). Positive τ is a compressive surface stress: the surface pushes outward, so a slab released from the bulk lattice constant expands along that axis, as Figure N4 shows directly. This is minus the continuum surface stress f , for which positive means tensile. The facets span -5.02 to +4.40 N/m and, more importantly, differ in kind. (100) straddles zero, its two in-plane axes carrying surface stress of opposite sign -- compressive along one, tensile along the other, so released it expands on one axis and contracts on the other -- which is the lattice-space origin of a transverse splitting. (111) collapses to a single point: its three-fold symmetry forces $\tau_{xx} = \tau_{yy}$ exactly, making it the natural control surface. Sign of $\tau_{xx} - \tau_{yy}$ depends on the in-plane axis assignment, which is recorded per surface in `tau_infinity.csv`. Values are read from that file, fitted by `fit_tau_infinity.py`, and are L2 (thickness-converged, 90/720 Ry).